

Chapter 1:

question 2)

momentum: velocity (mass)

$$\frac{[L]}{[t]} \times [kg]$$

$$P: [L][t][M]$$

Per Unit area per Unit time

$$\div // \frac{[L^2]}{[t]}$$

$$\therefore [L]^{-1}[M][t]^{-2}$$

$$SI: m, kg, s^2$$

Force: mass (acceleration)

$$[M] \times \frac{[L]}{[t]^2}$$

$$F: [M][L][t]^{-2}$$

Per Unit area

$$\div // [L]^2$$

$$\therefore [M][L]^{-1}[t]^{-2}$$

$$kg, m, s^2$$

1.13) velocity profile $\neq$ Stress component

$$a) v_x = by, v_y = 0, v_z = 0$$

$$v = \langle by, 0, 0 \rangle$$

$$|\nabla v_x\rangle = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial v_x}{\partial y} [by] = b$$

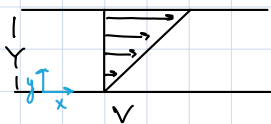
$$|\tau| = -\mu (|\nabla v| + |\nabla v|^T)$$

$$|\nabla v\rangle^T = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \dots & \dots \\ \frac{\partial v_x}{\partial z} & \dots & \dots \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\tau\rangle = \mu \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$p v \langle v | = \begin{bmatrix} p v_x v_x & p v_x v_y & p v_x v_z \\ p v_y v_x & p v_y v_y & p v_y v_z \\ p v_z v_x & p v_z v_y & p v_z v_z \end{bmatrix} = \begin{bmatrix} p(by)(by) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} p(b^2 y^2) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\langle v | \delta_x \rangle = by$$



$$b) v_x = by, v_y = bx, v_z = 0$$

$$|v\rangle = \langle by, bx, 0 \rangle \quad \langle x, y, z \rangle$$

$$|\nabla v\rangle = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\nabla v\rangle^T = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\tau\rangle = \mu \begin{bmatrix} 0 & b^2 & 0 \\ b^2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$p|v\rangle\langle v| = p \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_x v_y & v_y v_y & v_y v_z \\ v_x v_z & v_y v_z & v_z v_z \end{bmatrix} = p \begin{bmatrix} b_y b_y & (b_y)(b_x) & 0 \\ b_y(b_x) & b_x(b_x) & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ = p \begin{bmatrix} (b_y)^2 & b^2 x y & 0 \\ b^2 y x & (b_x)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$c) v_x = -b_y, v_y = b_x, v_z = 0$$

$$|\nabla|v\rangle = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\nabla|v\rangle^+ = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ -b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\tau\rangle = \mu \begin{bmatrix} 0 & -b+b & 0 \\ b-b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

$$p|v\rangle\langle v| = p \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_x v_y & v_y v_y & v_y v_z \\ v_x v_z & v_y v_z & v_z v_z \end{bmatrix} = p \begin{bmatrix} -b_y(-b_y) & -b_y(b_x) & 0 \\ -b_y(b_x) & b_x b_x & 0 \\ 0 & 0 & 0 \end{bmatrix} = p \begin{bmatrix} (b_y)^2 & -b^2 x y & 0 \\ -b^2 x y & (b_x)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$d) v_x = -\frac{1}{2}b_x, v_y = -\frac{1}{2}b_y, v_z = b_z$$

$$|\nabla|v\rangle = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} -\frac{b}{2} & 0 & 0 \\ 0 & -\frac{b}{2} & 0 \\ 0 & 0 & b \end{bmatrix}$$

$$|\tau\rangle = \mu \left[\begin{bmatrix} -\frac{b}{2} & -\frac{b}{2} \\ 0 & b \end{bmatrix} + \begin{bmatrix} -\frac{b}{2} & 0 \\ 0 & b \end{bmatrix} \right]$$

$$|\nabla|v\rangle^+ = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} -\frac{b}{2} & 0 & 0 \\ 0 & -\frac{b}{2} & 0 \\ 0 & 0 & b \end{bmatrix}$$

$$p|v\rangle\langle v| = p \begin{bmatrix} v_x v_x & v_y v_x & v_z v_x \\ v_y v_x & v_y v_y & v_y v_z \\ v_z v_x & v_y v_z & v_z v_z \end{bmatrix} = \begin{bmatrix} (\frac{1}{2}b_x)^2 & \frac{1}{4}b^2 x y & -\frac{1}{2}b^2 x z \\ \frac{1}{4}b^2 x y & \frac{1}{2}(b_y)^2 & -\frac{1}{2}b^2 y z \\ -\frac{1}{2}b^2 x z & -\frac{1}{2}b^2 y z & (b_z)^2 \end{bmatrix}$$

$$2.B2) \quad \frac{d\gamma_{xz}}{dx} = pg \cos(\beta)$$

$$\tau_{xz} = -\mu \frac{dv_z}{dx}$$

$$\frac{d(-\mu \frac{dv_z}{dx})}{dx} = pg \cos(\beta)$$

$$-\mu \frac{d^2 v_z}{dx^2} = pg \cos(\beta)$$

$$-\mu \int \frac{d^2 v_z}{dx^2} dx = \int pg \cos(\beta) dx$$

$$\frac{dv_z}{dx} = -\frac{pg \cos(\beta)x}{\mu} + c_1$$

$$v_z = -\frac{pg \cos(\beta)x^2}{2\mu} + c_1 x + c_2$$

$$v_z = -\frac{pg \cos(\beta)x^2}{2\mu} + 0(x) + \frac{pg \cos(\beta)\delta^2}{2\mu}$$

$$v_z = \frac{pg \cos(\beta)}{2\mu} (-x^2 + \delta^2)$$

$$B.C.1) \quad \text{at } x=0 \quad \tau_{xz}=0$$

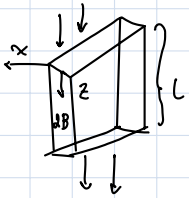
$$0 = -\frac{pg \cos(\beta)0}{\mu} + c_1 \\ c_1 = 0$$

$$B.C.2) \quad x=\delta \quad v_z=0$$

$$0 = -\frac{pg \cos(\beta)(\delta)^2}{2\mu} + 0(\delta) + c_2$$

$$c_2 = \frac{pg \cos(\beta)\delta^2}{2\mu}$$

2.B3)



Flow $B < r < L$
Assuming Steady State

$$0 = - \int_S ds p v_z < n > - \int_S ds p < s | n > + \int_V dv p < g >$$

a) from $B \ll W$, No Edge

Velocity has $x \neq z$ components

$$v = v_z(x) \text{ therefore } \tau = -\mu \left(\frac{\partial v_z}{\partial x} \right)$$

No slip with $B \rightarrow -B \quad 0 \Rightarrow zB$

$$0 = - \int_S ds < s | \tau | n > - \int_S ds p < s | n > + \int_V dv p < g >$$

$$\int_V dv p < g > = p g z = -p g z$$

$$p = p(z)$$

$$= \mu \left(\frac{\partial v_z}{\partial x} \right) + p + (-p g z)$$

$$\mu \left(\frac{\partial v_z}{\partial x} \right) = \frac{p_0 - p_L}{L}$$

$$\int \frac{\partial v_z}{\partial x} x dx = \int \frac{p_0 - p_L}{L} x dx$$

$$\tau_{zx} = \frac{p_0 - p_L}{L} x + C_1$$

$$-\mu \left(\frac{\partial v_z}{\partial x} \right) = \frac{p_0 - p_L}{L} x + C_1$$

$$\int \frac{\partial v_z}{\partial x} = \int \frac{p_0 - p_L}{\mu L} + \frac{C_1}{\mu}$$

$$v_z = \frac{p_0 - p_L}{2\mu L} x^2 + \frac{C_1}{\mu} x + C_2$$

$$\therefore \tau_{xz} = \frac{p_0 - p_L}{L} x$$

$$v_z = \frac{p_0 - p_L}{2\mu^2} B^2 \left(1 - \left(\frac{x}{B} \right)^2 \right)$$

$$b) v_z = \frac{p_0 - p_L}{2\mu L} B^2 \left(1 - \left(\frac{x}{B} \right)^2 \right)$$

w/ No Slip

$$v_{z \text{ max}} = \frac{p_0 - p_L}{2\mu L} B^2$$

$$\frac{v_z}{v_{\text{max}}} = 1 - \left(\frac{x}{B} \right)^2$$

$$\frac{\int_0^W \int_0^B 1 - \left(\frac{x}{B} \right)^2 dx dz}{\int_0^W \int_0^B 1 dx dz} = \frac{\int_0^1 1 - \left(\frac{x}{1} \right)^2 dx}{\int_0^1 dx}$$

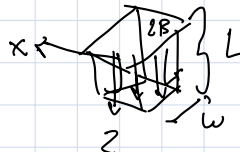
$$1 - 1/3 = 2/3$$

$$c) W = p < 2BW > \times v_z$$

$$p < 2BW > \times v_z = p < 2BW > \left(\frac{2}{3} \right) \left(\frac{p_0 - p_L}{2\mu L} \right) B^2$$

$$\frac{2}{3} \frac{(p_0 - p_L)}{\mu L} p B^3 W$$

d) when $W = B$ flow is not affected.
we can't base velocity on $v_z = v(x)$

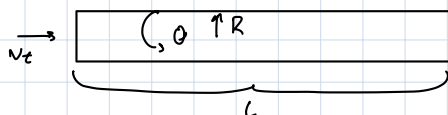


2.B.7)

$$\int_S ds \frac{\partial}{\partial t} p v_i = - \int_S ds p v < v | n > - \int_S ds < s | \tau | n > + \int_V dv p < g >$$

$$0 = - \int_S ds < s | \tau | n > + \int_V dv p < g >$$

Known velocity is $N_z = v_0$



Velocity changes as it approaches the wall. Velocity is a function of R

$$\tau_{zr} = -\mu \frac{\partial v_z}{\partial r}$$

$$\therefore \int_V dv p < g >$$

$$0 = - \int_S ds < s | \tau | n >$$

$$-\mu \frac{\partial v_z}{\partial r} = \frac{C_1}{r} \rightarrow \frac{\partial v_z}{\partial r} = \frac{C_1}{r} \left(-\frac{1}{r} \right)$$

$$v_z = -\frac{C_1}{2\mu} \ln r + C_2$$

Boundary conditions $N_z = v_0$

$$N_z \rightarrow r = 0$$

$$N_z / n R = v_0$$

$$\frac{v_z}{v_0} = -D \ln \frac{r}{R} + D_2$$

$$D_1 = 0 \quad D_2 = \frac{1}{\ln R}$$

$$\therefore \frac{v_z}{v_0} = \frac{\ln \frac{R}{r}}{\ln R}$$

$$b) W = \int_0^{2\pi} \int_0^R p v_z r dr d\theta$$

$$2\pi p \frac{v_0 R^2}{\ln R} \int_0^1 (\ln k) x dx$$

$$2\pi p \frac{v_0 R^2}{\ln R} \left(\frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 \right) \Big|_0^1$$

$$2\pi p \frac{v_0 R^2}{\ln R} \left(-\frac{1}{4} \ln 4 - \frac{1}{4} (1 - \ln 4) \right)$$

$$W = \frac{\pi p v_0 R^2}{\ln R} \left(\frac{1 - \ln 2}{\ln 2} - \frac{1}{2} \right)$$

$$c) F = \int_0^L \int_0^{2\pi} \mu \frac{\partial v_z}{\partial r} r d\theta dz$$

$$2\pi k L \mu V_0 \left(\frac{r}{h} \right)$$

$$F_z = - \frac{2\pi L \mu V_0}{\ln \frac{1}{\epsilon}}$$

$$D) F_z = - \frac{2\pi L \mu V_0}{\ln \frac{1}{\epsilon}} \quad k = 1 - \epsilon$$

$$F_z = - \frac{2\pi L \mu V_0}{\ln \left(\frac{1}{1-\epsilon} \right)}$$

taylor series inspection

$$f(0) + \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\therefore - \frac{2\pi L \mu V_0}{\epsilon} \left(1 - \frac{1}{2}\epsilon - \frac{1}{12}\epsilon^2 + \dots \right)$$

$$pgh + p|z=0 - p|z=L = (p_c - p)gh$$

$$V_z(r) = \frac{(p_c - p)gh}{4\mu L} R^2 + c_2 = 0$$

no slip conditions

$$c_2 = \frac{(p_c - p)gh}{4\mu L} R^2$$

$$V_z(r) = \frac{(p_c - p)gh}{4\mu L} (R^2 - r^2)$$

$$W = \frac{2\pi}{dt} = \frac{d(pV)}{dt} = p \frac{dV}{dt}$$

$$W = pV_2 \pi R^2$$

$$W = p \int V_z(2\pi r dr) \Rightarrow 2\pi p \int_0^R r V_z dr$$

$$2\pi p \int_0^R r \frac{p_c - p}{4\mu L} (R^2 - r^2) dr$$

$$W = \frac{\pi p (p_c - p) g h R^4}{8\mu L}$$

plug & solve

$$W = 3.9 \times 10^5 \frac{\text{J}}{\text{hr}}$$

2. B8)

$$V_z = V(r)$$

$$\int_S ds p V_z \frac{\partial}{\partial x} = - \int_S ds p V_z \langle v|n \rangle - \int_S ds \langle \delta|t|n \rangle - \int_S ds p \langle \delta|n \rangle + \int_V dv p \langle q \rangle$$

$$(2\pi r \Delta r) \theta_{zz}|_{z=0} - (2\pi r \Delta r \theta_{zz})|_{z=L} + 2\pi r L \theta_{zz}|_L - 2\pi (r + \Delta r) \theta_{zz}|_{r+\Delta r} p g \sin \theta = 0$$

$$\div 2\pi \Delta r L$$

$$0 = -r \frac{\theta_{zz}|_{z=L} - \theta_{zz}|_{z=0}}{L} - \frac{(r + \Delta r) \theta_{rz}|_{r+\Delta r} - r \theta_{rz}|_r}{\Delta r} + p g r \sin \theta$$

$$-r \frac{\theta_{zz}|_{z=L} - \theta_{zz}|_{z=0}}{L} - \frac{d}{dr} (r \theta_{rz}) + p g r \sin \theta = 0$$

$$\theta_{zz} = p \theta_{zz} + \theta_{zz} + p v_z v_z = p + p v_z^2$$

$$v \text{ prof. } -r \frac{p|z=0 - p|z=L}{L} - \frac{d}{dr} (r \tau_{rz}) = 0$$

$$\sin \theta \Rightarrow h = L \sin \theta$$

$$\sin \theta = \frac{h}{L}$$

$$\frac{d}{dr} (r \tau_{rz}) = \frac{pgh + p|z=0 - p|z=L}{L}$$

$$\frac{d}{dr} \left(-\mu \frac{dv_z}{dr} \right) = \frac{pgh + p|z=0 - p|z=L}{L}$$

$$V_z = - \frac{pgh + p|z=0 - p|z=L}{L} R^2 - c_2 = 0$$